Christopher P. Higgins, Civil and Environmental Engineering

James F. Ranville, Chemistry

John McCray, Civil and Environmental Engineering

John R. Spear, Civil and Environmental Engineering

Jonathan Sharp, Civil and Environmental Engineering

Linda A. Figueroa, Civil and Environmental Engineering

Thomas Monecke, Geology and Geological Engineering

Zhaoshan Chang, Geology and Geological Engineering

Associate Professors

Mathias Burisch, Geology and Geological Engineering

Yvette Kuiper, Geology and Geological Engineering

Assistant Professor

Ryan Venturelli, Geology and Geological Engineering

Affiliate Faculty

Katharina Pfaff, Geology and Geological Engineering, Research Associate Professor

Emeriti Professor

John B. Curtis, Geology and Geological Engineering

Wendy J. Harrison, Geology and Geological Engineering

Donald L. Macalady, Chemistry

Patrick MacCarthy, Chemistry

Samuel B. Romberger, Geology and Geological Engineering

Richard F. Wendlandt, Geology and Geological Engineering

Thomas R. Wildeman, Chemistry

Emeriti Associate Professor

L. Graham Closs, Geology and Geological Engineering

E. Craig Simmons, Chemistry

GIS & GeoInformatics

Program Description

The interdisciplinary online program in Geographic Information Systems and GeoInformatics (GIS & GeoInformatics) focuses on the applications of geospatial technology. It provides hands-on geospatial training, multi-criteria decision-making, advanced applications, and quantitative analysis in GIS and Remote Sensing (RS). This program aligns directly with the Colorado School of Mines' strengths in Earth, Energy, and Environment. Our programs enhance students' quantitative geospatial analytical skills, position them ahead of the technology curve, and empower professionals to advance their careers.

Student Learning Outcomes

Students are expected to know and be able to do the following by the time of graduation.

- Master the fundamental concepts of GIS and the skills necessary for conducting geospatial analyses using GIS and/or RS.
- Master the essential principles of optical remote sensing and apply them in image processing.
- Learn and apply the key concepts and principles of Synthetic Aperture Radar (SAR) and Interferometric SAR (InSAR) in InSAR processing.
- Conduct digital elevation model (DEM) acquisition and preparation, creating new map products on a GIS platform to detect spatial and temporal variations of phenomena, such as geomorphic features, landslides, flooding, or forest fires.
- Identify topographic features through the superimposed Digital Elevation Model (DEM) and DEM-derived map products.
- Develop custom Python-based geoprocessing tools and enhance the existing capabilities of GIS applications through Python scripting.
- Automate geoprocessing tasks and#complex spatial#workflows via Python scripting.
- Design and conduct GIS-based projects integrating various datasets, geospatial knowledge, and analytical skills.

Primary Contact

Program Director: Wendy Zhou
Program Coordinator: Rachel McDonald

https://gis.mines.edu/

Certificate and Degree Requirements

We offer a graduate certificate and a non-thesis Master of Science degree. The courses taken for the certificate degree can be used towards the Master's degree. These programs are available as an online program.

The Master of Science (Non-Thesis) Program in GIS & GeoInformatics

The Master of Science (Non-Thesis) Program outlined below may be completed by individuals already holding an undergraduate or advanced degree or as a combined degree program by individuals already matriculated as undergraduate students at Colorado School of Mines. Courses taken while working on any of the four GIS & GeoInformatics graduate certificates can be applied to this Master of Science program. The program is comprised of a total of 30 credit hours.

All Master of Science (Non-Thesis) program will include the following core requirements:

GEGN575	APPLICATIONS OF GEOGRAPHIC INFORMATION SYSTEMS	3.0
GEGN590	GIS-BASED REAL WORLD LEARNING PROJECT I - FUNDAMENTALS	3.0
GEGN592	GIS-BASED REAL WORLD LEARNING PROJECT II - ADVANCED APPLICATIONS	3.0

And, seven of the following courses:

GEGN542	ADVANCED DIGITAL TERRAIN ANALYSIS	3.0
GEGN579	PYTHON SCRIPTING FOR GEOGRAPHIC INFORMATION SYSTEMS	3.0

GEGN580	APPLIED REMOTE SENSING FOR GEOENGINEERING AND GEOSCIENCES	3.0
GEGN588	ADVANCED PLANETARY GEOGRAPHIC INFORMATION SYSTEMS	3.0
GEOL557	EARTH RESOURCE DATA SCIENCE 1: FUNDAMENTALS	3.0
GEOL558	EARTH RESOURCE DATA SCIENCE 2: APPLICATIONS AND MACHINE-LEARNING	3.0
MNGN502	GEOSPATIAL BIG DATA ANALYTICS	3.0
MNGN557	MINERAL ECONOMICS AND POLICY	2.0
MNGN561	PROJECT MANAGEMENT	3.0
MNGN562	MINING ENVIRONMENTAL AND SOCIAL RESPONSIBILITY	2.0
DSCI503	ADVANCED DATA SCIENCE	3.0
DSCI570	INTRODUCTION TO MACHINE LEARNING	3.0
MATH530	INTRODUCTION TO STATISTICAL METHODS	3.0
CEEN501	LIFE CYCLE ASSESSMENT	3.0
Students attending on campus also may select from semester based, in person courses GEGN532, GEGN568, CEEN581.		

Mines Combined Undergraduate / Graduate Program

Students enrolled in Mines' combined undergraduate/graduate program may double count up to six credits of graduate coursework to fulfill requirements of both their undergraduate and graduate degree programs. These courses must have been passed with "B-" or better, not be substitutes for required coursework, and meet all other University, Department, and Program requirements for graduate credit.

Students are advised to consult with their undergraduate and graduate advisors for appropriate courses to double count upon admission to the combined program.

The Graduate Certificate Program in GIS & GeoInformatics outlined below may be completed by individuals already holding an undergraduate or advanced degree or as a combined degree program by individuals already matriculated as graduate students at Colorado School of Mines. The graduate certificate is comprised of:

Course Work	12.0
Total Semester Hrs	**12.0**

Graduate Certificate of GIS & GeoInformatics:

Students working towards a Graduate Certificate of GIS & GeoInformatics are required to take any four of the following courses:

One Required Course:		
GEGN575	APPLICATIONS OF GEOGRAPHIC INFORMATION SYSTEMS	3.0
And, any three of the following courses:		
GEGN542	ADVANCED DIGITAL TERRAIN ANALYSIS	3.0
GEGN579	PYTHON SCRIPTING FOR GEOGRAPHIC INFORMATION SYSTEMS	3.0
GEGN580	APPLIED REMOTE SENSING FOR GEOENGINEERING AND GEOSCIENCES	3.0
GEGN588	ADVANCED PLANETARY GEOGRAPHIC INFORMATION SYSTEMS	3.0
GEOL557	EARTH RESOURCE DATA SCIENCE 1: FUNDAMENTALS	3.0
GEOL558	EARTH RESOURCE DATA SCIENCE 2: APPLICATIONS AND MACHINE-LEARNING	3.0
CEEN501	LIFE CYCLE ASSESSMENT	3.0
DSCI503	ADVANCED DATA SCIENCE	3.0
DSCI570	INTRODUCTION TO MACHINE LEARNING	
MATH530	INTRODUCTION TO STATISTICAL METHODS	3.0
MNGN502	GEOSPATIAL BIG DATA ANALYTICS	3.0
MNGN557	MINERAL ECONOMICS AND POLICY	2.0
MNGN561	PROJECT MANAGEMENT	3.0
MNGN562	MINING ENVIRONMENTAL AND SOCIAL RESPONSIBILITY	2.0
Students attending on campus also may select from semester based, in person courses GEGN532, GEGN568, CEEN581.		

Program Director

Wendy Zhou, Professor, Geology & Geological Engineering

Program Associate Director

Sebnem Duzgun, Professor, Fred Banfield Endowed Chair in Mining Engineering

Professors

Sebnem Duzgun, Professor, Fred Banfield Endowed Chair in Mining Engineering

Amy Landis, Civil and Environmental Engineering, Professor, Mines Presidential Faculty Fellow for Access, Attainment & Diversity

Wendy Zhou, Professor, Geology & Geological Engineering

Teaching Professor

Wendy Fisher, Computer Science, Teaching Professor

Research Professor

Zane Jobe, Research Associate Professor, Chevron Center of Research Excellence Director, Geology & Geological Engineering

Humanitarian Engineering and Science

Program Description

The MS degrees in Humanitarian Engineering and Science (HES) are a professional MS (non-thesis) and a thesis-based MS. These degrees are targeted to recent graduates or mid-career professionals with a BS in science and engineering who are interested in careers, research opportunities, and/or acquiring skills that will help them work effectively with communities. The degrees include a core HES curriculum plus an approved track of related courses in a science or engineering discipline.

The HES graduate certificate is designed for professionals seeking to attend school part time or students who are seeking degrees in other